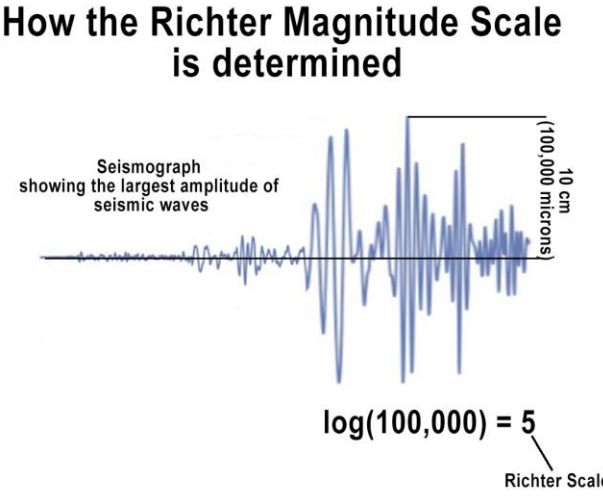
EART11200: Natural Scientist's Toolkit





The Richter scale uses logarithm to base 10

EARTHQUAKE MAGNITUDE

Magnitude
Seismic Wave Energy

10
9
8
7
6
5
4
3
2

Magnitude is a number that characterizes the relative size of an earthquake. Each whole number increase represents a tenfold increase in the measured amplitude, and **32 times more energy release**.

Examples of major earthquakes and their magnitudes:

Here are some factors that determine how destructive an earthquake can be:

Location · Magnitude · Depth · Distance from epicenter · Geologic conditions · Secondary effects · Architecture

Sichuan 7.9 2008

Tōhoku 9.1 2011

Kashmir 7.6 2005

Kobe 6.9 1995

Sumatra 2004 9.1

3 of the 5 largest earthquakes in the 21st century have occurred in the Sumatra region of Indonesia.

The resulting tsunami from this earthquake was recorded around the world, and killed more people than any other in recorded history. Over 230,000 people are missing and presumed dead.

Christchurch 2011 6.2

The 2011 Christchurch earthquake was both shallow and very close to the city, resulting in extensive damage. In contrast, a 7.1 magnitude earthquake a year prior caused much less damage as it was deeper and located further from population centers.

Alaska 1964 9.2

San Francisco 1989 6.9

Many people in Haiti live and work in unreinforced buildings, so the 2010 earthquake caused extensive damage.

Haiti 7.0 2010

Valdivia 1960 9.4

This earthquake off the Chilean coast – the most powerful ever recorded – left two million people homeless, killed approximately 1,655, and injured thousands more.

< 1

Energy release of world's largest nuclear test

3

Energy release of Mount St. Helens eruption

18
Loss of life, $1 billion+ in damage

Energy release equivalent to the Hiroshima atomic bomb

150
Moderate shaking, property damage

1,500
Light shaking, some property damage

Energy release of an average tornado

10,000
Felt by humans

Energy release of a large lightning bolt

100,000

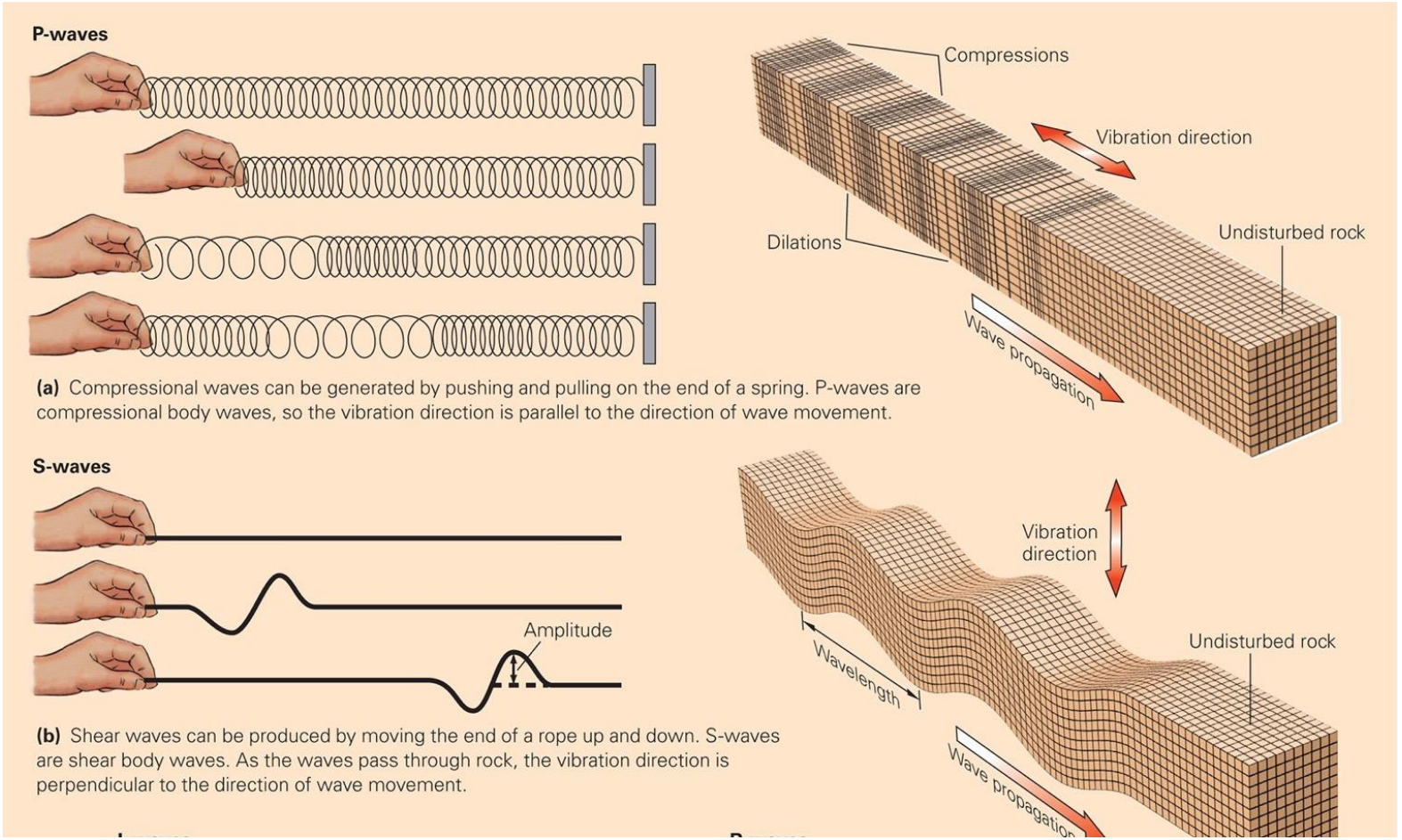
1,000,000

Number of earthquakes per year, globally

Source: U.S. Geological Survey

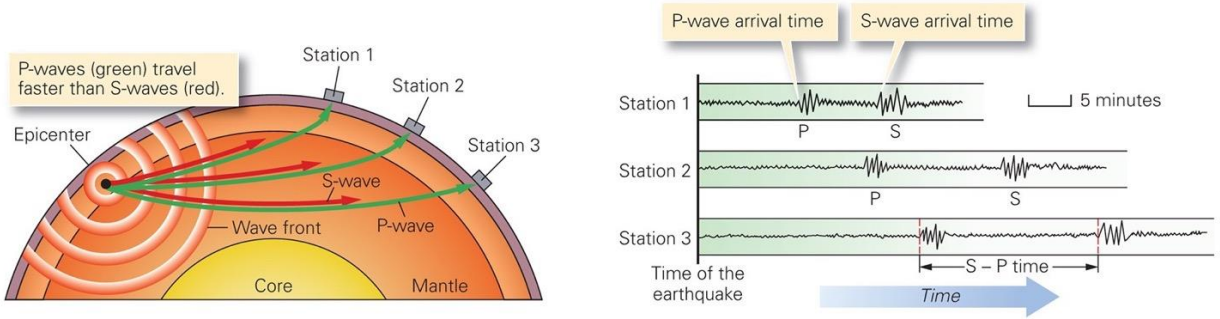
Richter scale intro

Seismic waves – see Portrait of a Planet 339

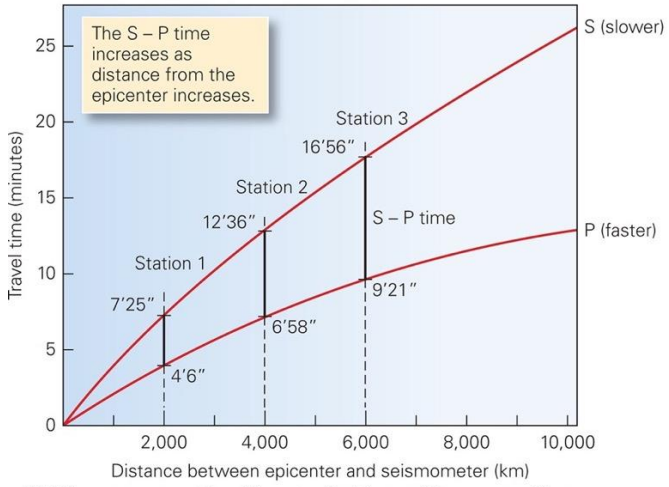


The Richter scale

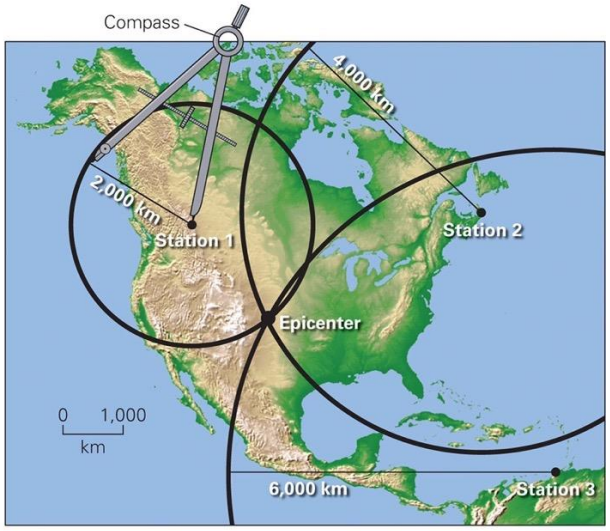
background - see Portrait of a Planet 339



(a) Different types of seismic waves travel at different velocities. The greater the distance between the epicenter and the seismometer station, the longer it takes for earthquake waves to arrive, and the greater the delay between the P-wave and S-wave arrival times.



(b) We can represent the different arrival times of P-waves and S-waves on a graph of travel-time curves.



(c) If an earthquake epicenter lies 2,000 km from Station 1, we draw a circle with a radius of 2,000 km around the station at the scale of the map. We do the same for the other two stations. The intersection of the three circles is the epicenter.

Problem 3.7 *The Richter Scale is one way to quantify the magnitude of Earthquakes. The magnitude of an Earthquake is independent of distance from the Earthquake as it refers to the magnitude of the Earthquake at the epicentre.*

$$m_l = \log_{10}\left(\frac{A}{A_0}\right)$$

where A_0 is the amplitude of the a wave at the epicentre that would just be detected (mm) and A is the amplitude of the wave measured (mm).
A_0 is given by

$$-\log_{10}(A_0) \cong 1.4 + 0.16\sqrt{D}$$

where D is the distance to the epicentre in km.
In this exercise, you will convert the amplitudes produced by an Earthquake at distance D to an equivalent magnitude for if the Earthquake occurred locally (where $D = 0$).

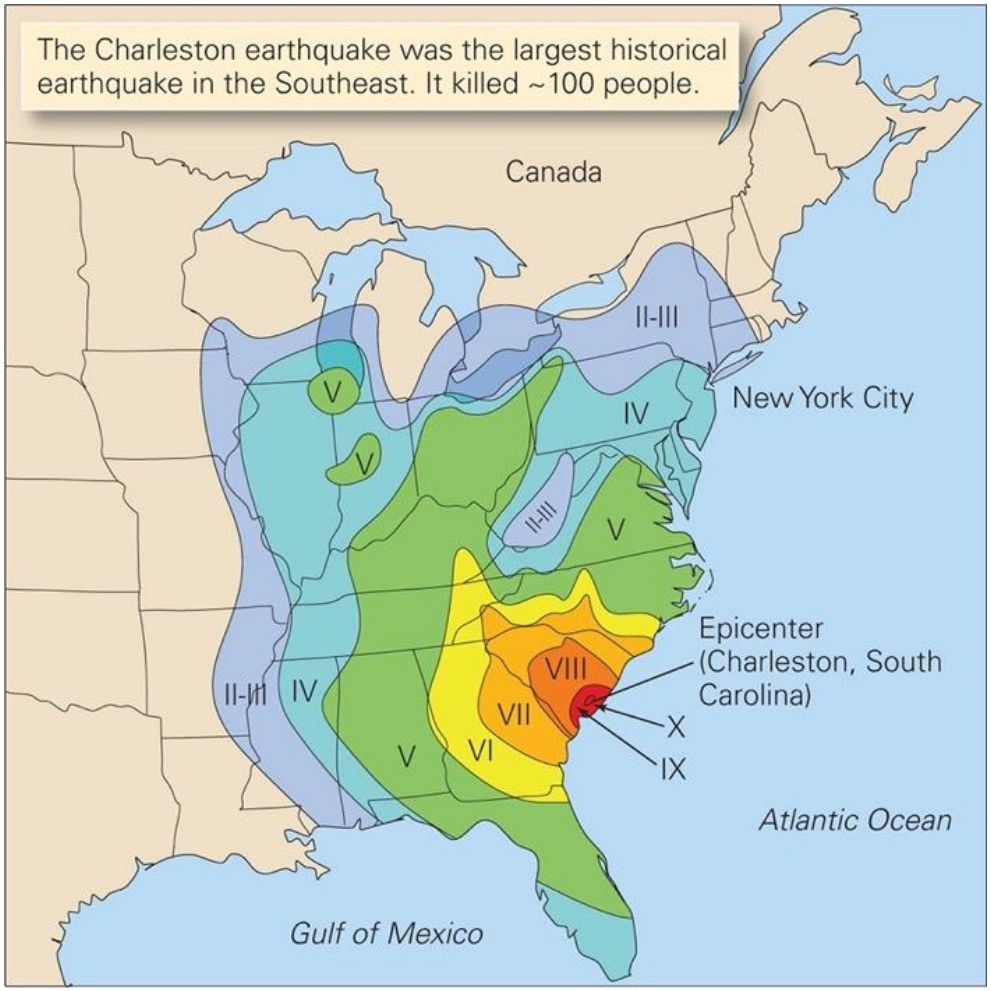
The 1886 Earthquake in Charlston, South Carolina is estimated to be a 7.3 on the Richer magnitude scale. Complete the analysis below.

1. *Calculate the amplitude, A, measured $D = 100$ km away from the epicentre.*
2. *If this amplitude was measured at the epicentre, $D = 0$ km, what would the magnitude be?*
3. *Calculate the amplitude measured $D = 1000$ km away in New York City.*
4. *If this amplitude in New York City was measured at the epicentre of an Earthquake what would the magnitude be ?*

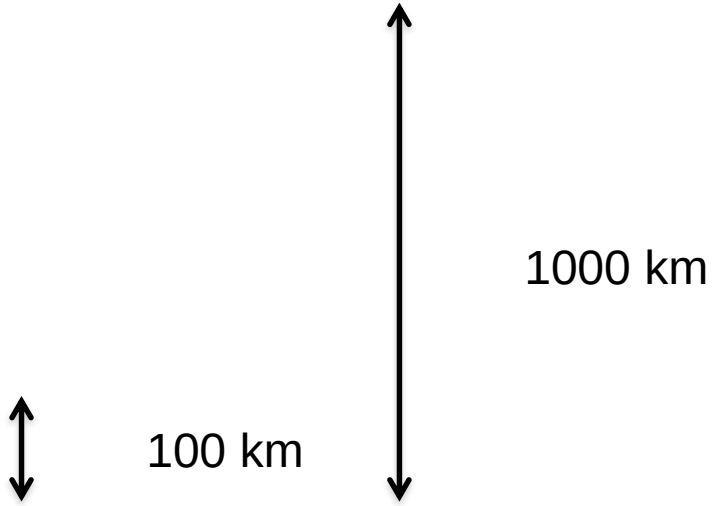
The Richter scale problem

Charleston Earthquake, magnitude 7.3

see Portrait of a Planet 339



Compare local magnitude with the table overleaf



Magnitude	Description	Mercalli intensity	Average earthquake effects	Average frequency of occurrence globally (estimated)
1.0–1.9	Micro	I	Microearthquakes, not felt, or felt rarely. Recorded by seismographs.[30]	Continual/several million per year
2.0–2.9	Minor	I to II	Felt slightly by some people. No damage to buildings.	Over one million per year
3.0–3.9		III to IV	Often felt by people, but very rarely causes damage. Shaking of indoor objects can be noticeable.	Over 100,000 per year
4.0–4.9	Light	IV to VI	Noticeable shaking of indoor objects and rattling noises. Felt by most people in the affected area. Slightly felt outside. Generally causes none to minimal damage. Moderate to significant damage very unlikely. Some objects may fall off shelves or be knocked over.	10,000 to 15,000 per year
5.0–5.9	Moderate	VI to VII	Can cause damage of varying severity to poorly constructed buildings. At most, none to slight damage to all other buildings. Felt by everyone.	1,000 to 1,500 per year
6.0–6.9	Strong	VIII to X	Damage to a moderate number of well-built structures in populated areas. Earthquake-resistant structures survive with slight to moderate damage. Poorly designed structures receive moderate to severe damage. Felt in wider areas; up to hundreds of miles/kilometers from the epicenter. Strong to violent shaking in epicentral area.	100 to 150 per year
7.0–7.9	Major	X or greater[31]	Causes damage to most buildings, some to partially or completely collapse or receive severe damage. Well-designed structures are likely to receive damage. Felt across great distances with major damage mostly limited to 250 km from epicenter.	10 to 20 per year
8.0–8.9	Great		Major damage to buildings, structures likely to be destroyed. Will cause moderate to heavy damage to sturdy or earthquake-resistant buildings. Damaging in large areas. Felt in extremely large regions.	One per year
9.0 and greater			At or near total destruction – severe damage or collapse to all buildings. Heavy damage and shaking extends to distant locations. Permanent changes in ground topography.	One per 10 to 50 years

(*Based on U.S. Geological Survey documents.*)[32]

Quantifying earthquakes

- We have discussed the Richter scale, which measures the amplitude of an earthquake.
- The Mercalli scale measures (qualitative) destructiveness or effect.
- The energy released (intensity) of the earthquake is given by

$$\text{Log}_{10} E = 11.8 + 1.5M$$

Where M is magnitude on the Richter scale